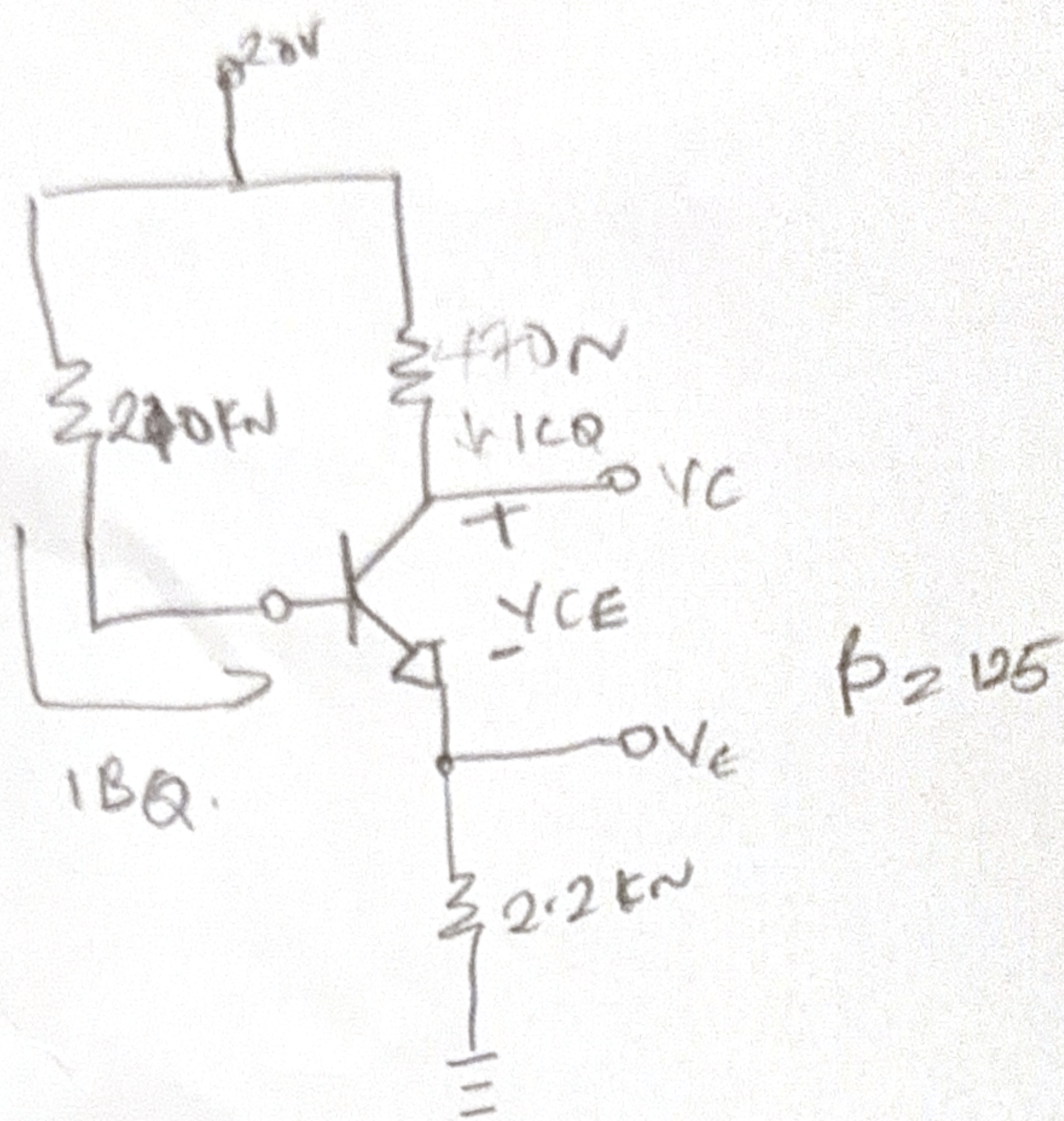


Name: Naimul Yekini

Determine I_B , I_C , V_{CE} , V_C , V_B and V_E .



$$I_C = \beta I_B$$

$$I_E = I_C + I_B$$

$$I_E = (\beta + 1) I_B$$

$$I_E = I_C$$

$$V_{BE} = 0.7 \text{ V}$$

$$\beta = 125$$

$$V_{CC} = 20 \text{ V}$$

$$R_C = 470 \Omega$$

$$R_E = 2.2 \text{ k}\Omega$$

$$\text{For } I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$I_E = (\beta + 1) I_B, \quad V_E = I_E R_E \quad V_B = V_E + V_{BE}$$

$$I_B = \frac{V_{CC} - (V_{BE} + (\beta + 1) I_B R_E)}{R_B}$$

$$I_B = \frac{20 - 0.7}{270 + 126 \times 2.2} = \frac{19.3}{547.2} = 3.53 \times 10^{-5} \text{ A}$$

$$② I_C = \beta I_B = 125(35.27) = 4.409 \text{ mA}$$

②

$$I_E = (\beta + 1) I_B = 125(35.27) = 4.44 \text{ mA}$$

3) Mode voltage -

$$V_E = I_E R_E = (4.44 \text{ mA})(2.2 \text{ k}\Omega)$$

$$= 9.7774$$

$$V_B = V_E - 0.7 = 10.4774$$

$$V_C = V_{CC} - I_C R_C = 20 - (4.409)(470)$$

$$= 17.9284$$

$$4) V_{CE} = V_C - V_E$$

$$= 17.928 - 9.7777$$

$$= 8.154$$